Chapter 1 Vector

FSE10023 Linear Algebra



Prepared by: Chai Ching Yi

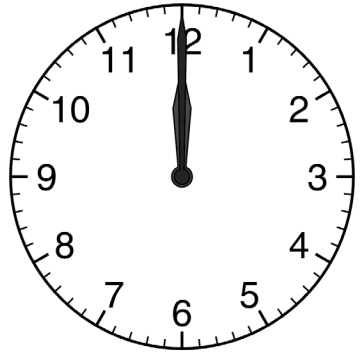
Learning Outcomes

1. Demonstrate knowledge of the properties of vectors and apply appropriate vector operations to solve practical problems.
2. Demonstrate knowledge of the relation between the Cartesian co-ordinate system and Polar co-ordinate system to transpose functions between the two coordinate systems.
3. Describe the locus graphically in Cartesian co-ordinates and polar co-ordinates.
4. Describe complex numbers graphically and solve problems using the operations on a+bi form, polar form and De Moivre's theorem.
5. Execute basic mathematical operations involving matrices.

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Scalar Quantities





- Quantities than can be defined by its magnitude.
- Scalar = number
- Example: Time, mass, distance, length, temperature, energy, volume.
- Purpose: to describe physical quantities **without direction**.
- Example:
    a. The distance between two tables is 1 meter.
    b. The duration of this class is 2 hours.

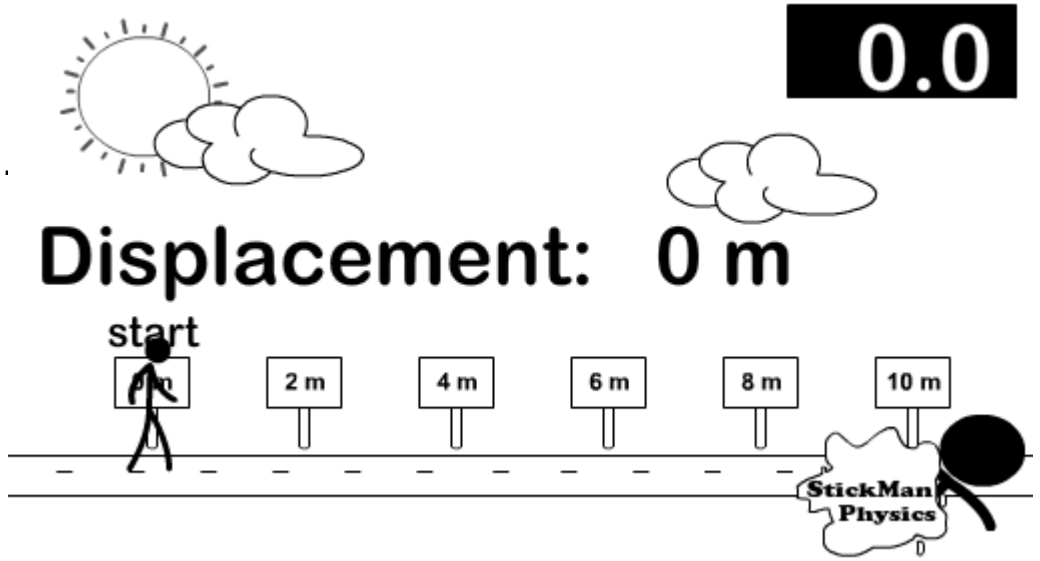


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Vector Quantities

- Quantities than can be defined by its magnitude and **direction**.
- Example: displacement, velocity, force.
- Purpose: to describe physical quantities **with direction**.
- Example:



  a. In a rescue mission, the rescue team must know the distance to the distress signal, and the direction from which the signal is coming from.

  b. I would like to go to the nearest McDonalds as soon as possible. So, I seek help from Google Map. Google Map will guide me with the shortest distance, direction and time required.

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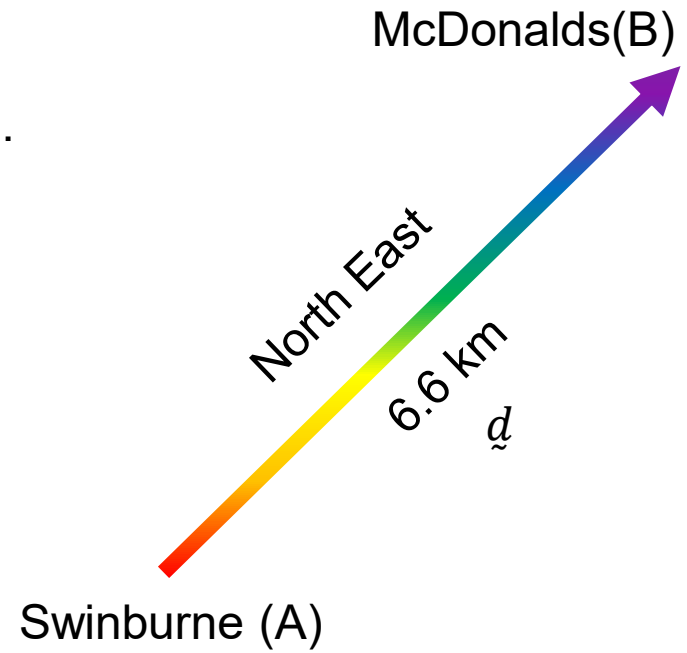
Vector Representation and Notation

For example: I want to go to nearest McDonalds from Swinburne.

- Distance is 6.6km, d = 6.6km (= scalar quantity)
- Distance is 6.6km to the North East is, $\underset{\sim}{d}$ (= vector quantity)

General graphical representation:

- Magnitude = length of line
- Direction = arrowhead $(\overrightarrow{AB})$



Vector Notation: $\overrightarrow{AB}, AB, \mathrm{a}, \underset{\sim}{a}$

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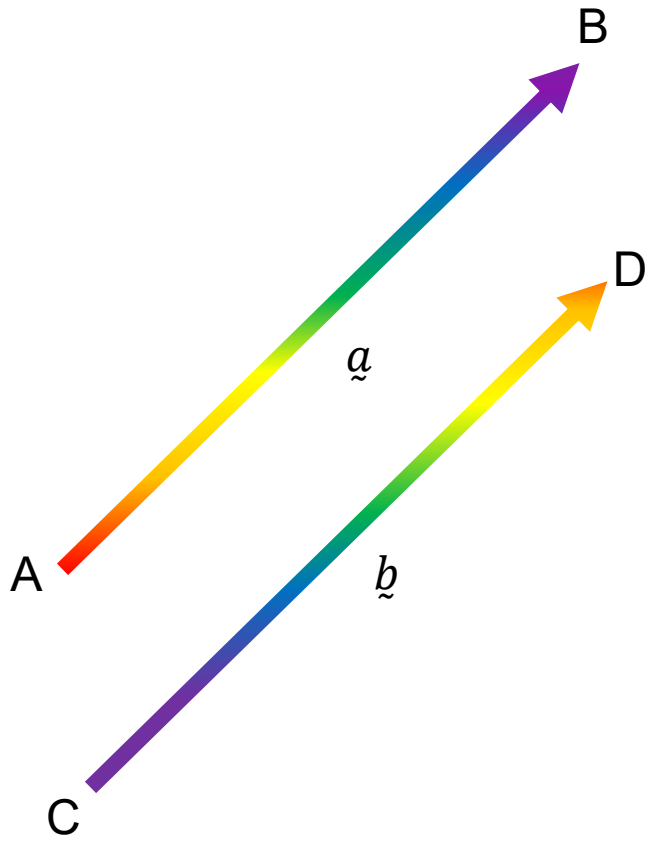
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Equal Vector

Two vectors are equal if both have **same** magnitude (length) and direction.

For example:

- $\underset{\sim}{a} = \underset{\sim}{b}$
- $\overrightarrow{AB} = \overrightarrow{CD}$



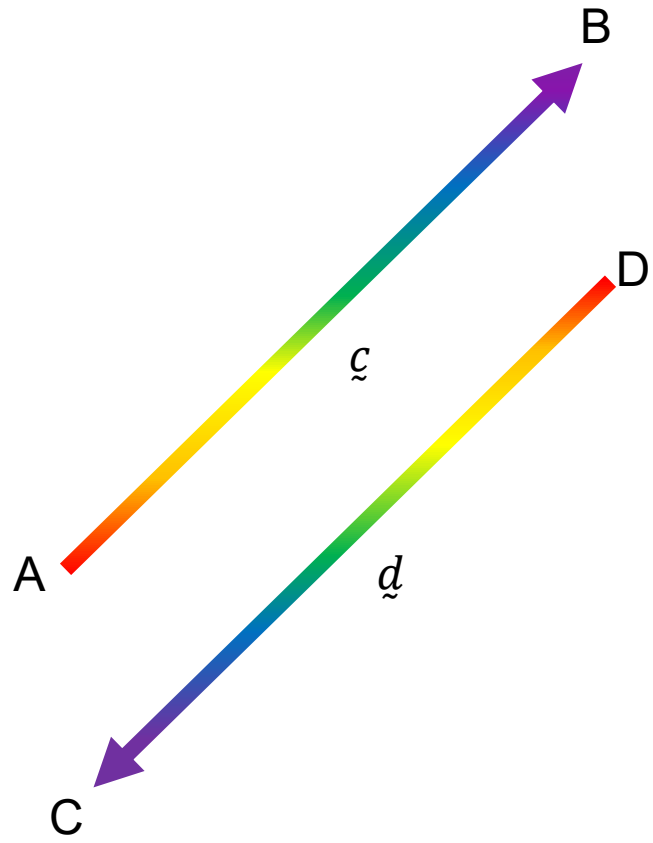
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Negative Vector

Two vectors are negative if both have **same** magnitude (length) and **opposite** direction.

For example:

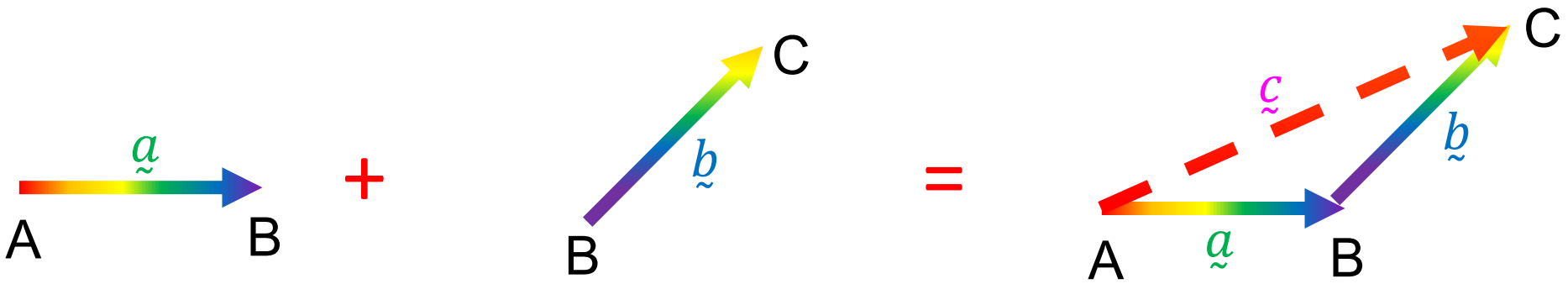
- $\underset{\sim}{c} = -\underset{\sim}{d}$
- $\overrightarrow{AB} = \overrightarrow{DC}$



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Vector Addition: Graphical Method

The top-to-tail method

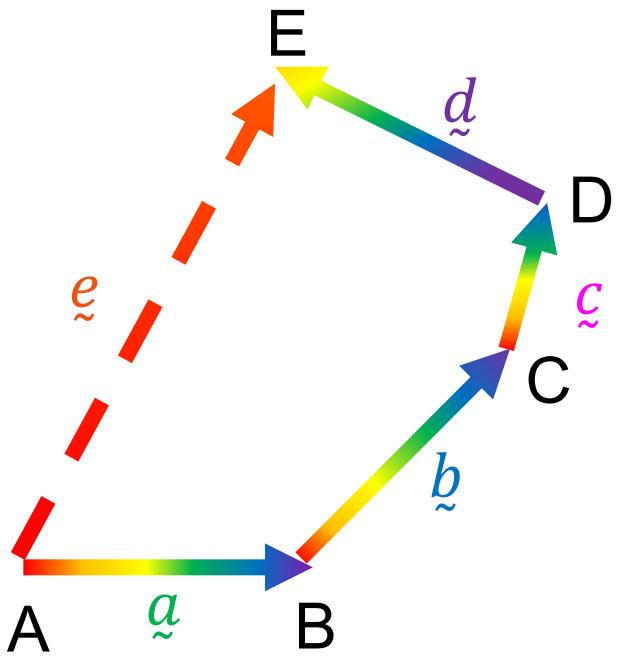


$\underset{\sim}{a} + \underset{\sim}{b} = \underset{\sim}{c}$ (Resultant vector)

$\overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AC}$

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Vector Addition: Graphical Method



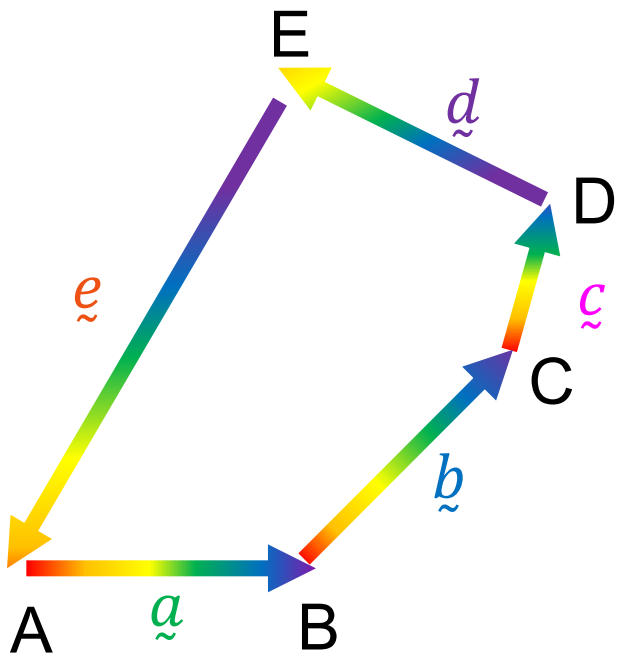
$\underset{\sim}{a} + \underset{\sim}{b} + \underset{\sim}{c} + \underset{\sim}{d} = \underset{\sim}{e}$ (Resultant vector)

$\overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CD} + \overrightarrow{DE} = \overrightarrow{AE}$

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Vector Addition: Zero Vector



$$\underset{\sim}{a} + \underset{\sim}{b} + \underset{\sim}{c} + \underset{\sim}{d} + \underset{\sim}{e} = 0$$

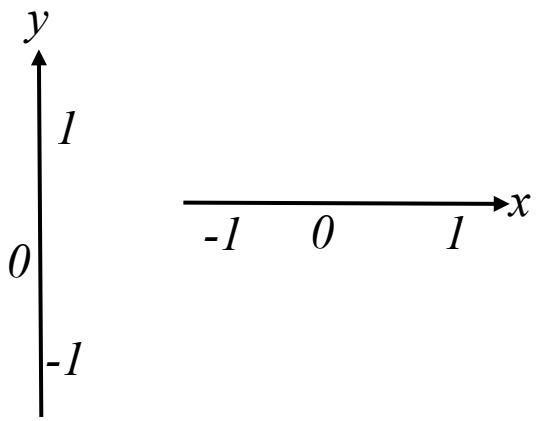
$$\overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CD} + \overrightarrow{DE} + \overrightarrow{EA} = 0$$

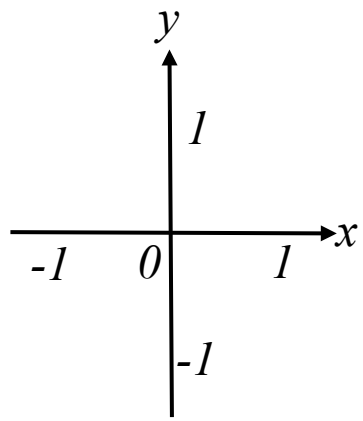
Zero vector = vector with a magnitude of zero (0).

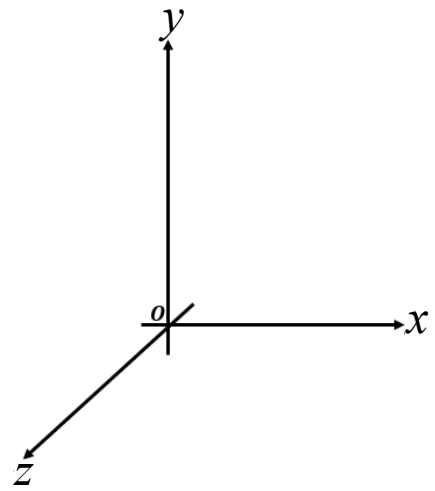
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Dimensional Space (1D, 2D, 3D)







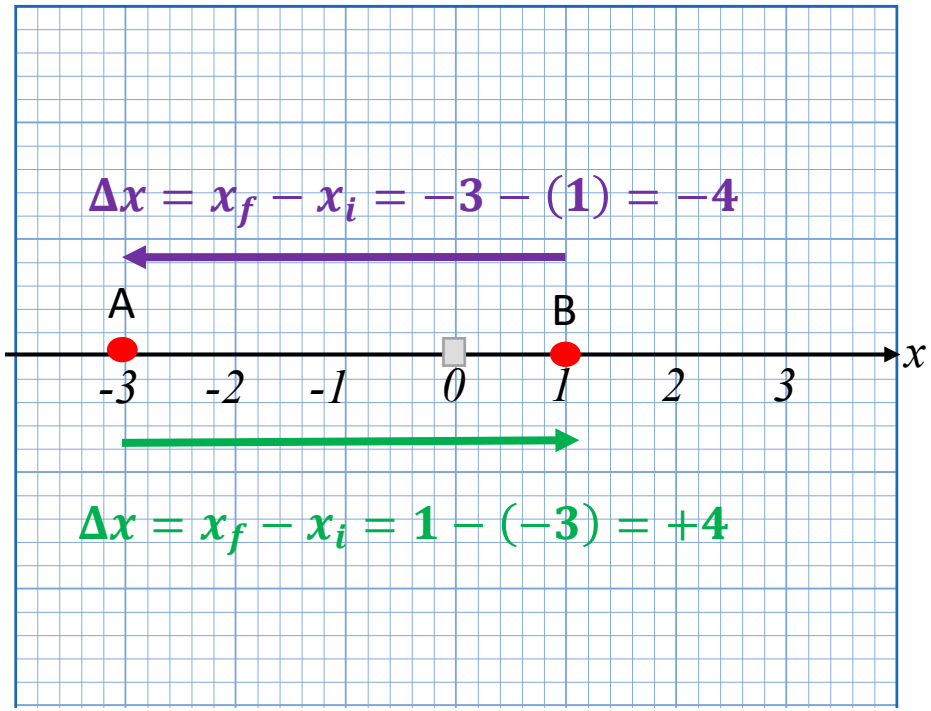
1 Dimension (1D)

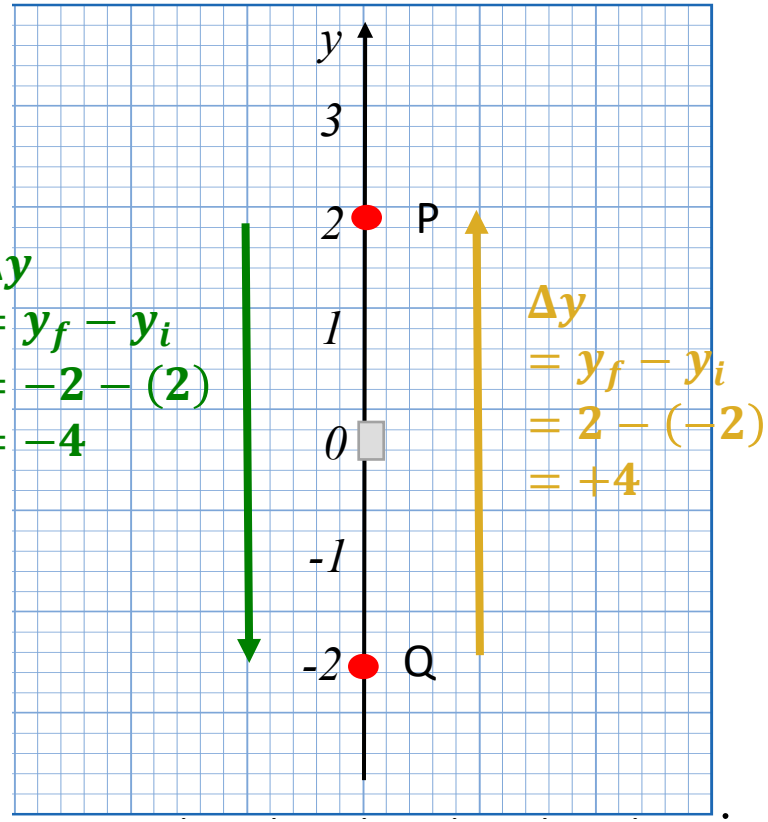
2 Dimension (2D)

3 Dimension (3D)

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1 Dimension: Displacement

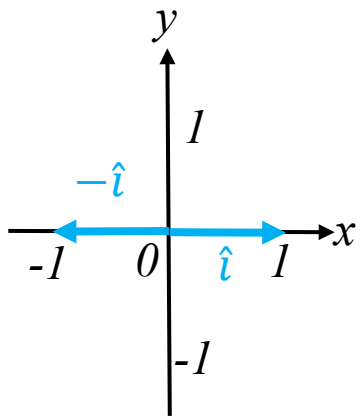


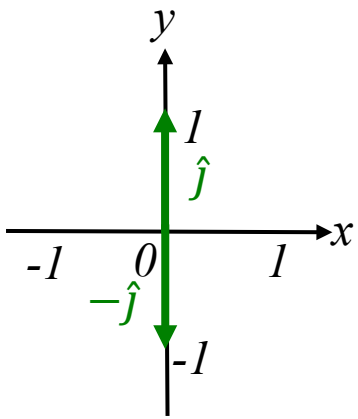


Displacement = Final point (*f*) – Initial point (*i*)

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2D and Unit Vectors



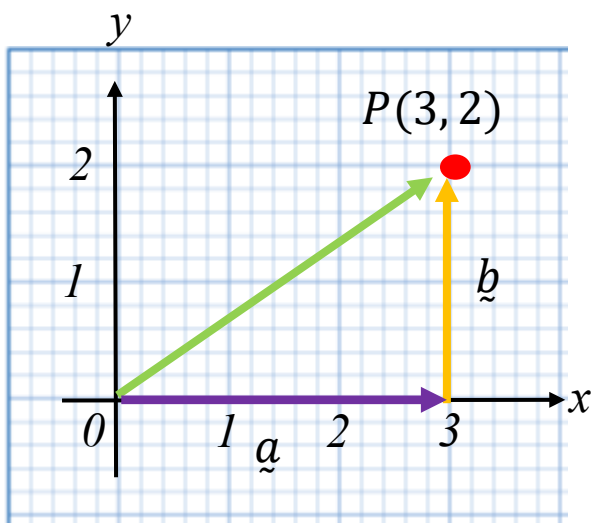


- A *unit vector* is a vector that has a *magnitude of one*.
- Purpose: to show direction.
- In a 2D Cartesian Plan, unit vectors are:
  - *a)* $\hat{\imath}$: unit vector in the positive x-axis direction.
  - *b)* $-\hat{\imath}$: unit vector in the negative x-axis direction.
  - *c)* $\hat{\jmath}$: unit vector in the positive y-axis direction.
  - *d)* $-\hat{\jmath}$: unit vector in the negative y-axis direction.

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Position Vector



- Position vector $\overrightarrow{OP}$: Position of Point *P (3,2)* with respect to the origin *0 (0,0)*.
- $\overrightarrow{OP}$ is defined by two components, $\vec{a}$ (along the x-axis) and $\vec{b}$ along the y-axis.
- $\underset{\sim}{a} = 3\hat{\imath} = \begin{pmatrix} 3 \\ 0 \end{pmatrix}$; $\underset{\sim}{b} = 2\hat{\jmath} = \begin{pmatrix} 0 \\ 2 \end{pmatrix}$
- $\overrightarrow{OP} = \underset{\sim}{a} + \underset{\sim}{b}$

$$= \begin{pmatrix} 3 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 2 \end{pmatrix}$$

$$= \begin{pmatrix} 3 \\ 2 \end{pmatrix}$$

$$= 3\hat{\imath} + 2\hat{\jmath}$$

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Position Vectors Presentation

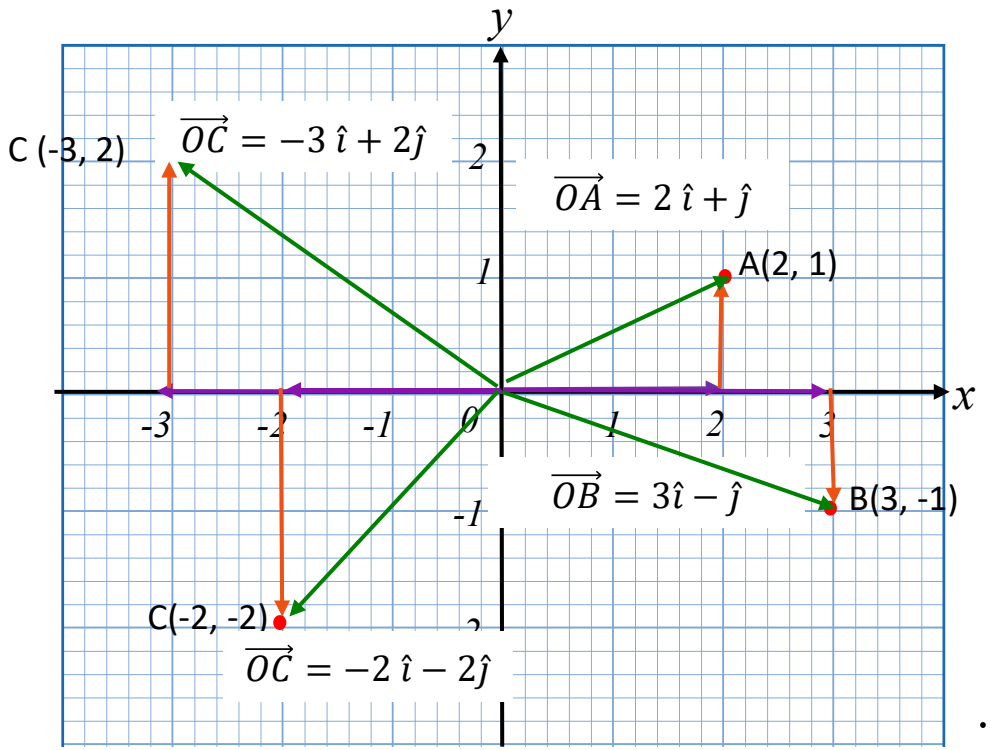
- Position Vector can be presented in two ways:
    1. Unit Vector Notation (UV)
    2. Magnitude and Direction Notation (MD)



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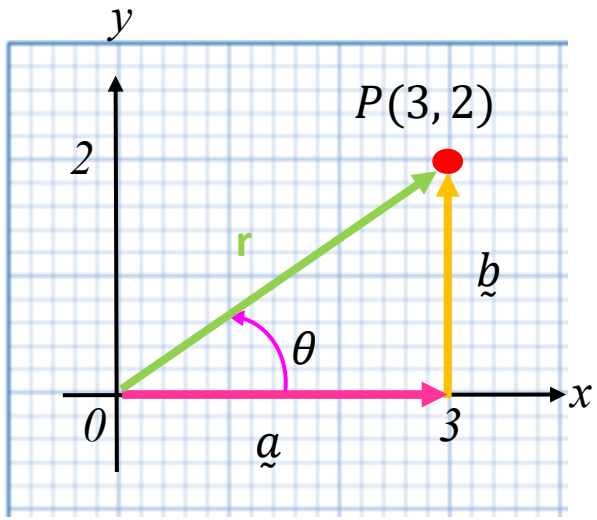
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Position Vectors: 1. UV Notation



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Position Vectors: 2. MD Notation (1st quadrant)



- By Unit Vector Notation, $\overrightarrow{OP} = 3\hat{i} + 2\hat{j}$
- $|\underset{\sim}{a}| = 3$; $|\underset{\sim}{b}| = 2$
- Applying Pythagoras Theorem:

$$\left|\overrightarrow{OP}\right| = r = \sqrt{a^2 + b^2} = \sqrt{3^2 + 2^2} = \sqrt{13}$$

- Direction of vector $\overrightarrow{OP}$ $(\angle\theta)$

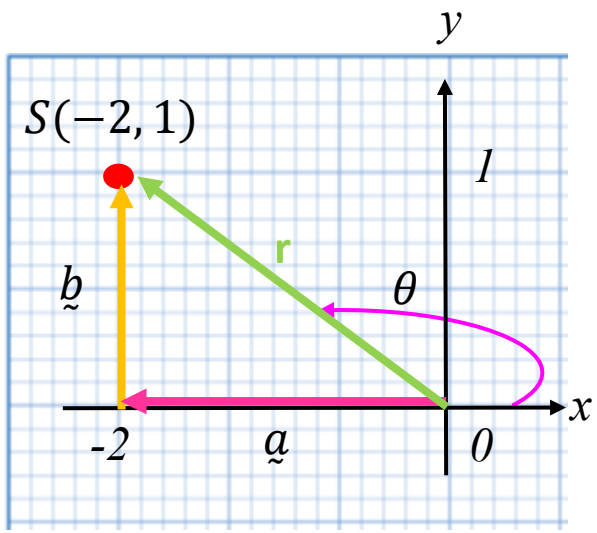
$$\theta = \tan^{-1}\left|\frac{2}{3}\right| = 33.69°$$

$$\therefore \overrightarrow{OP} = r\angle\theta = \sqrt{13}\angle 33.69°$$

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Position Vectors: 2. MD Notation (2nd quadrant)



- By Unit Vector Notation, $\overrightarrow{OS} = -2\hat{i} + \hat{j}$
- $|\underset{\sim}{a}| = -2$; $|\underset{\sim}{b}| = 1$
- Applying Pythagoras Theorem:

$$|\overrightarrow{OS}| = r = \sqrt{(-2)^2+(1)^2} = \sqrt{5}$$

- Direction of vector $\overrightarrow{OS}$ $(\angle\theta)$

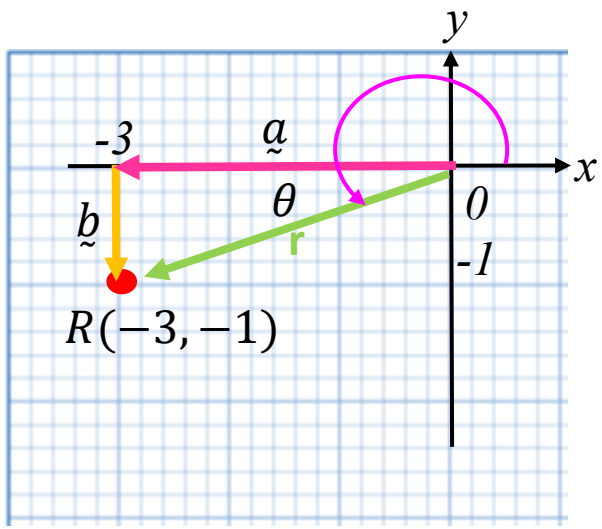
$$\theta = 180° - \tan^{-1}\left|\frac{1}{2}\right| = 153.43°$$

$\therefore \overrightarrow{OS} = \sqrt{5}\angle 153.43°$

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Position Vectors: 2. MD Notation (3rd quadrant)



- By Unit Vector Notation, $\overrightarrow{OR} = -3\hat{i} - \hat{j}$
- $|\underset{\sim}{a}| = -3$; $|\underset{\sim}{b}| = -1$
- Applying Pythagoras Theorem:

$$\left|\overrightarrow{OR}\right| = r = \sqrt{(-3)^2+(-1)^2} = \sqrt{10}$$

- Direction of vector $\overrightarrow{OR}$ $(\angle\theta)$

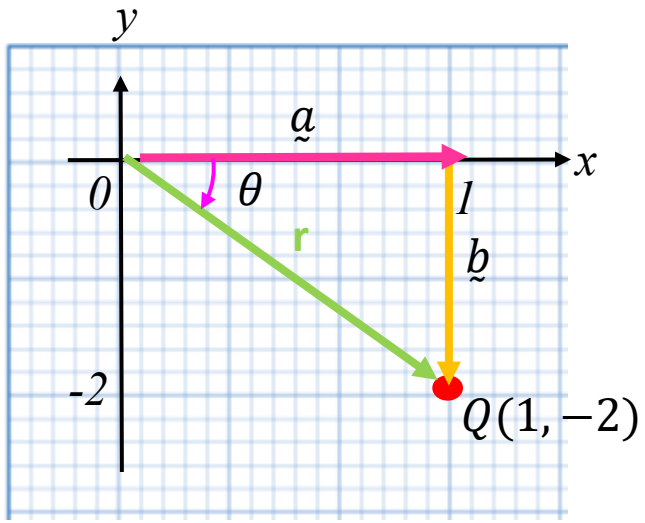
$$\theta = 180° + \tan^{-1}\left|\frac{-1}{-3}\right| = 198.43°$$

$$\therefore \overrightarrow{OR} = \sqrt{10}\angle 198.43°$$

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Position Vectors: 2. MD Notation (4th quadrant)



- By Unit Vector Notation, $\overrightarrow{OQ} = \hat{i} - 2\hat{j}$
- $|\underset{\sim}{a}| = 1$; $|\underset{\sim}{b}| = -2$
- Applying Pythagoras Theorem:

$$\left|\overrightarrow{OQ}\right| = r = \sqrt{1^2 + (-2)^2} = \sqrt{5}$$

- Direction of vector $\overrightarrow{OQ}$ $(\angle\theta)$

$$\theta = \tan^{-1}\left|\frac{-2}{1}\right| = -63.43°$$

$$\therefore \overrightarrow{OQ} = \sqrt{5}\angle -63.43°$$

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Example 1: 2D Position Vectors

For the following four points in 2D Cartesian Plane, write the position vectors in:

a) Unit Vector notation

b) Magnitude and direction

i. A (4, 3)
ii. B (-10, 12)
iii. C (-5, -8)
iv. D (6, -9)



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Example: 2D Position Vectors

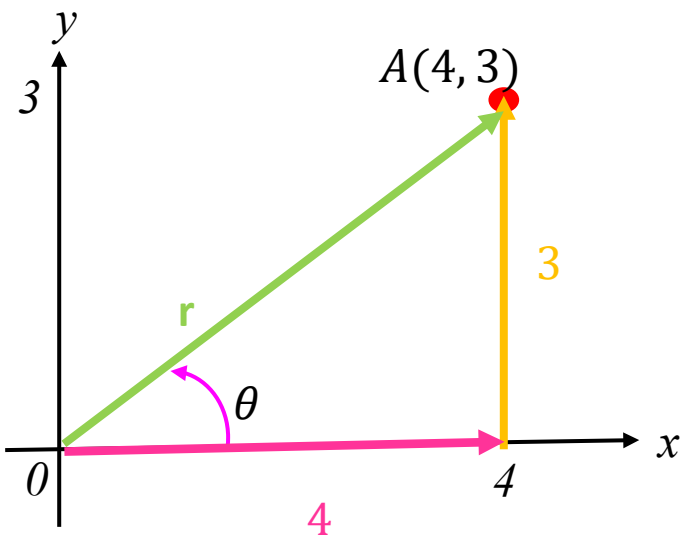
i. A (4, 3)

a) UV: $\overrightarrow{OA} = 4\,\hat{\imath} + 3\hat{\jmath}$

b) MD: $\theta = \tan^{-1}\left|\frac{3}{4}\right| = 36.87°$

$r = \sqrt{(4)^2+(3)^2} = 5$

$\overrightarrow{OA} = 5\angle 36.87°$



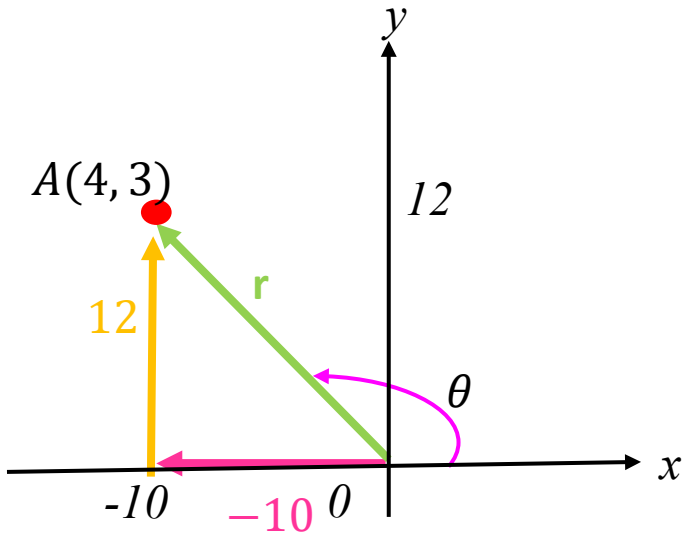
ii. B (-10, 12)

a) $\overrightarrow{OB} = -10\,\hat{\imath} + 12\hat{\jmath}$

b) $\theta = 180° - \tan^{-1}\left|\frac{12}{10}\right| = 129.81°$

$r = \sqrt{(-10)^2+(12)^2} = 2\sqrt{61}$

$\overrightarrow{OB} = 2\sqrt{61}\angle 129.81°$



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Example: 2D Position Vectors

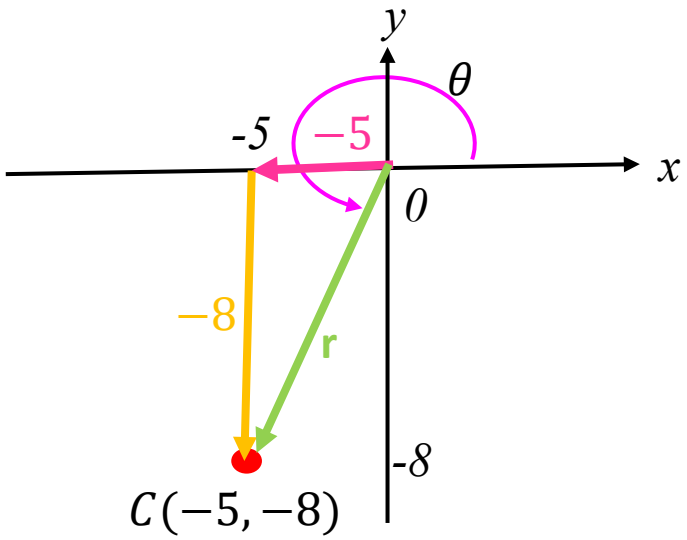
iii. C (-5, -8)

a) $\overrightarrow{OC} = -5\,\hat{\imath} - 8\hat{\jmath}$

b) $\theta = 180° + \tan^{-1}\left|\frac{8}{5}\right| = 237.99°$

$r = \sqrt{(-5)^2+(-8)^2} = \sqrt{89}$

$\overrightarrow{OC} = \sqrt{89}\angle 237.99°$



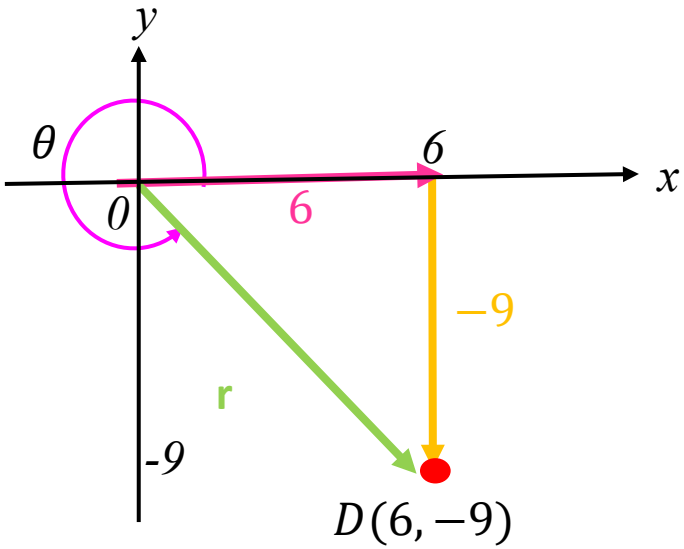
iv. D (6, -9)

a) $\overrightarrow{OD} = 6\,\hat{\imath} - 9\hat{\jmath}$

b) $\theta = 360° - \tan^{-1}\left|\frac{9}{6}\right| = 303.69°\ or\ -56.31°$

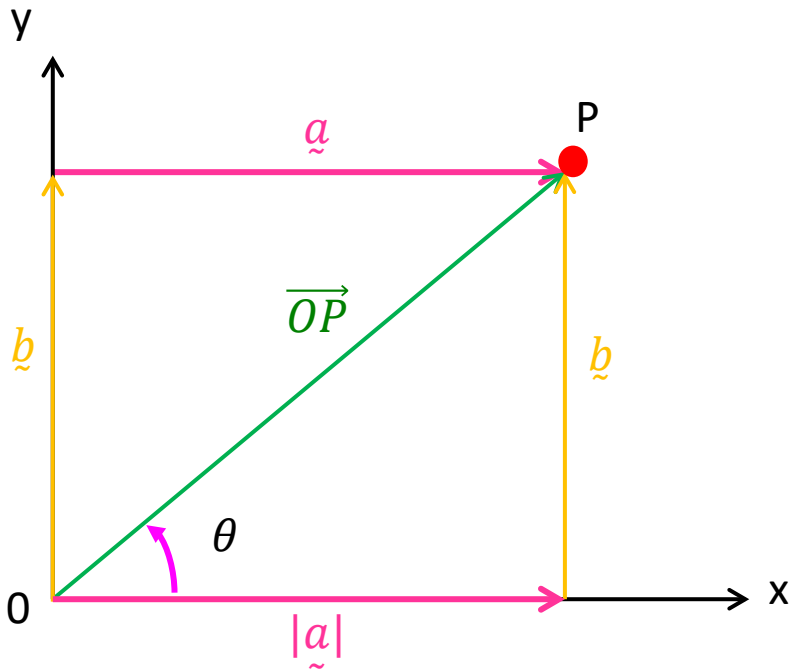
$r = \sqrt{(6)^2+(-9)^2} = 3\sqrt{13}$

$\overrightarrow{OD} = 3\sqrt{13}\angle 303.69°$ or $3\sqrt{13}\angle -56.31°$



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Conversion: Magnitude and Direction to Unit Vector



$$\overrightarrow{OP} = (r\cos\theta)\hat{\imath} + (r\sin\theta)\hat{\jmath}$$

- UV presentation: $\overrightarrow{OP} = \underset{\sim}{a} + \underset{\sim}{b}$, we need to find the magnitude $\underset{\sim}{a}$ and $\underset{\sim}{b}$.
- If the magnitude of $|\overrightarrow{OP}| = r$ & direction $= \angle\theta$, we know that:

$$\cos\theta = \frac{a}{r}$$

$$a = r\cos\theta$$

$$|\underset{\sim}{a}| = a = r\cos\theta$$

$$\sin\theta = \frac{b}{r}$$

$$b = r\sin\theta$$

$$|\underset{\sim}{b}| = b = r\sin\theta$$

$$\boldsymbol{\overrightarrow{OP} = \underset{\sim}{a} + \underset{\sim}{b} = (r\cos\theta)\hat{\imath} + (r\sin\theta)\hat{\jmath}}$$

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Example 2: 2D Position Vectors Conversions

Express the following vectors in unit vectors notation:

i. $\overrightarrow{OP} = 10\angle 60°$

ii. $\overrightarrow{OR} = 6\angle -120°$

iii. $\overrightarrow{OS} = 5\angle \frac{5}{6}\pi$

iv. $\overrightarrow{OQ} = 3\angle -\frac{\pi}{6}$



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Example 2: 2D Position Vectors Conversions

i. $\overrightarrow{OP} = 10\angle 60°$

$= 10\left[(\cos 60)\hat{\imath} + (\sin 60)\hat{\jmath}\right]$

$= 5\hat{\imath} + 5\sqrt{3}\hat{\jmath}$

ii. $\overrightarrow{OR} = 6\angle -120°$

$= 6\left[(\cos(-120))\hat{\imath} + (\sin(-120))\hat{\jmath}\right]$

$= -3\hat{\imath} - 3\sqrt{3}\hat{\jmath}$

iii. $\overrightarrow{OS} = 5\angle \frac{5}{6}\pi$

$= 5\left[\left(\cos\left(\frac{5}{6}\pi\right)\right)\hat{\imath} + \left(\sin\left(\frac{5}{6}\pi\right)\right)\hat{\jmath}\right]$

$= \frac{-5\sqrt{3}}{2}\hat{\imath} + \frac{5}{2}\hat{\jmath}$

iv. $\overrightarrow{OQ} = 3\angle -\frac{\pi}{6}$

$= 3\left[\left(\cos\left(-\frac{\pi}{6}\right)\right)\hat{\imath} + \left(\sin -\frac{\pi}{6}\right)\hat{\jmath}\right]$

$= \frac{3\sqrt{3}}{2}\hat{\imath} - \frac{3}{2}\hat{\jmath}$

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Week 1: To Do in Tutorial Class

- Exercise 1: All Questions
- Exercise 2: Question 1 to 6



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That’s all for today.

Any questions?



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